

Full-production spectral discretization to continuum PDE: P3 stages 1-2

TECT verification-first repository · claim A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM

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1. Purpose and scope

The P1 backend supplies a discrete variational functional and its real gradient. P2 proves well-posedness of the corresponding continuum full-production gradient flow on a fixed three-torus. Neither result by itself proves that an N32, N64, or N128 nonlinear solver trajectory approximates that continuum flow. P3 is the separate discretization-to-continuum package.

This v1.0 note freezes the numerical objects and closes only the first spatial baseline. Finite-time flow convergence, Hessian/Ritz convergence, and hardware/precision agreement remain open.

1 Continuum and discrete objects

Let $R(\Psi) = \nabla_{\mathbb{R}} F(\Psi)$ denote the real- L^2 gradient of the P1 reference functional, in the eta-shell-zero scope of P2. Let S_N be the three-component trigonometric polynomial space represented by the cubic FFT grid and let P_N be the real- L^2 Fourier projector.

The exact spectral Galerkin vector field is

$$R_N^G(\Psi_N) = P_N R(\Psi_N).$$

The hash-pinned backend instead differentiates the periodic collocation energy; call the result R_N^C . Spectral differentiation is exact on S_N , but pointwise nonlinear products need not remain in S_N . Therefore $R_N^C = R_N^G$ is not assumed. The measured aliasing/consistency quantity is

$$e_N^R = \frac{\|R_N^C - P_N R_{N_{\text{ref}}}^C\|_{L^2}}{\max(\|P_N R_{N_{\text{ref}}}^C\|_{L^2}, 10^{-30})}.$$

Two same-backend reference grids are used, and their projected difference is reported separately. This prevents a single under-resolved reference from defining its own apparent convergence. It does not provide an independent implementation of the continuum Euler–Lagrange residual; that remains a load-bearing closure item.

For the collocation functional F_N^C , the discrete real pairing is

$$\langle u, v \rangle_{N, \mathbb{R}} = \Delta V \operatorname{Re} \sum_{a, x} \overline{u_a(x)} v_a(x).$$

The energy-gradient audit verifies a centred finite-difference form of $DF_N^C(\Psi_N)v = \langle R_N^C(\Psi_N), v \rangle_{N, \mathbb{R}}$.

2 Manufactured spatial field

Each complex component is a nonzero constant plus several low Fourier modes with distinct phases and directions. The construction keeps the density far above the positive rho floor and activates every pinned Class-II coefficient. It is analytic, so spectral convergence is expected rather than a fixed low-order finite-difference rate. The acceptance criterion asks for at least fourth algebraic order over the declared grid sequence; the measured rate is also retained rather than replaced by the threshold.

3 Executed result

The grids are $N = 8, 12, 16$ and the same-backend oversampling references are $N = 24, 32$. The result is:

N	e_N^R	e_N^{ref}	e_N^F	e_N^{DF}
8	$5.3301196 \cdot 10^{-6}$	$5.5885294 \cdot 10^{-16}$	$1.7564784 \cdot 10^{-13}$	$1.0210053 \cdot 10^{-8}$
12	$4.2482236 \cdot 10^{-10}$	$3.6469415 \cdot 10^{-15}$	0	$3.9197285 \cdot 10^{-9}$
16	$6.6678548 \cdot 10^{-12}$	$2.3350092 \cdot 10^{-14}$	$2.2533398 \cdot 10^{-16}$	$3.9197706 \cdot 10^{-9}$

The two observed residual orders are 23.2750 and 14.4408. All 13 executable assertions pass. Energy monotonicity is interpreted above the declared 10^{-14} relative roundoff floor; values below that floor are machine-level agreement and are not ordered artificially.

4 Stage 2 finite-time result

Temporal and spatial errors are separated. On N8, the manufactured trajectory uses the semidiscrete forcing $S_N(t) = \partial_t \Psi_N^* + R_N^C(\Psi_N^*)$, so the experiment measures the classical RK4 time error rather than spatial aliasing. At final time 0.2, steps 0.04, 0.02, 0.01 give relative errors $5.6334 \cdot 10^{-11}$, $3.4821 \cdot 10^{-12}$, and $2.1644 \cdot 10^{-13}$, hence observed orders 4.0160 and 4.0079. These errors are in the truncation regime rather than at machine zero.

Separately, the unforced full-production gradient flow is integrated to time 0.005 with step $5 \cdot 10^{-4}$. N8, N12, and N16 are compared with projected N20 and N24 trajectories. Relative errors against N24 are $1.3063 \cdot 10^{-8}$, $1.0239 \cdot 10^{-12}$, and $1.3118 \cdot 10^{-14}$, with observed rates 23.3164 and 15.1465. The energy decreases at every recorded step on all five grids. This is same-backend trajectory self-convergence, not yet an independent continuum flow implementation.

5 What remains for closure

The following are load-bearing and are not inferred from the spatial table:

1. a forced manufactured finite-time solution with separated temporal and spatial errors, plus an unforced cross-grid trajectory check;
2. a matrix-free real Hessian audit on nested Fourier spaces, with Ritz residuals, a spectral gap, and an isolated-cluster statement;
3. CPU complex128 as the reference, CPU complex64 comparison, and CUDA complex64/complex128 comparison when the installed device supports them;
4. a one-command aggregate, immutable evidence hashes, independent operator reproduction, and an integrated convergence proof.

6 Devil’s-advocate review

First, Fourier differentiation alone does not remove nonlinear aliasing; this objection is valid and is mitigated by keeping R_N^C and R_N^G distinct. Second, one smooth manufactured field does not give a uniform theorem for all P2 solutions; this objection is upheld, and the claim remains T3. Third, energy convergence does not imply eigenvalue convergence; the isolated Ritz-cluster gate remains open. Fourth, a missing CUDA device cannot be reported as a cross-device pass; the hardware matrix must distinguish PASS, FAIL, and UNAVAILABLE.

7 Mandatory adversarial code review

Sign. Fourier derivatives use $+ik_j$ and the Laplacian uses $-|k|^2$; the spatial audit does not change the canonical backend signs. **Factor and convention.** FFT coefficients are divided by the grid-point count, inverse reconstruction multiplies by the target count, and the real pairing includes the cell volume. Centered crops use same-parity grids, so no Nyquist half-weight is silently introduced. **Units.** All values use the pinned dimensionless P1 normalization and the fixed length-16 torus; the table does not claim a dimensional laboratory error. **Convergence.** Two reference grids and three target grids are retained, with the reference projection uncertainty reported independently of the measured target error. The time-order rerun was deliberately moved away from machine precision after the first otherwise-passing run exposed roundoff-sensitive terminal errors. **Hardcode masking.** No measured residual, rate, or energy value is an input to the executable. Manufactured Fourier coefficients are declared test inputs; acceptance thresholds live in the manifest. **Limit cases.** A nonzero-density analytic field is intentional for this baseline and does not test removal of the rho floor, rough H2 data at time zero, or a Nyquist-loaded field. Those cases cannot be inferred from this table. External review of the Fourier normalization, same-backend-reference limitation, and later manufactured forcing is explicitly invited.

8 Reproduction

Run `python codes/foundations/a3_full_production_spatial_consistency.py`. The expected verdict is 13/13 PASS and A3-FULL-SPATIAL-CONSISTENCY-PASS.

Result footer

Result ID:	A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM
Precise statement:	Stages 1-2 define the Fourier consistency metrics and pass spatial and finite-time self-convergence audits.
Scope:	Fixed T3, pinned full-production parameters and floors, eta_shell=0, CPU complex128, analytic manufactured data.
Dependencies:	A1-PRODUCTION-FUNCTIONAL-REALISATION; A2-FULL-PRODUCTION-WELLPOSED; A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL.
Evidence grade:	ANALYTIC + EXECUTED + CONDITIONAL; T3 scaffold.
Reproduction command:	python codes/foundations/ a3_full_production_spatial_consistency.py python codes/foundations/ a3_full_production_finite_time_convergence.py
Expected output:	13/13 spatial PASS and 10/10 finite-time PASS.
Falsification gate:	A3-FULL-DISCRETIZATION-CLOSURE.
Tier before / after:	new / T3 COMPUTABLE-SCAFFOLD.
No-overclaim statement:	Same-backend self-convergence only; not an independent continuum residual, Hessian/Ritz, GPU/precision, Sector-B continuum certificate, T5, T6, or T7.
Next required action:	Hessian/Ritz isolated-cluster audit, then hardware and precision matrix and independent continuum residual.